

Energy_Efficiency Performance Report 2022 (ID: doc-syn-20)

1. Executive Summary & Production Efficiency Analysis

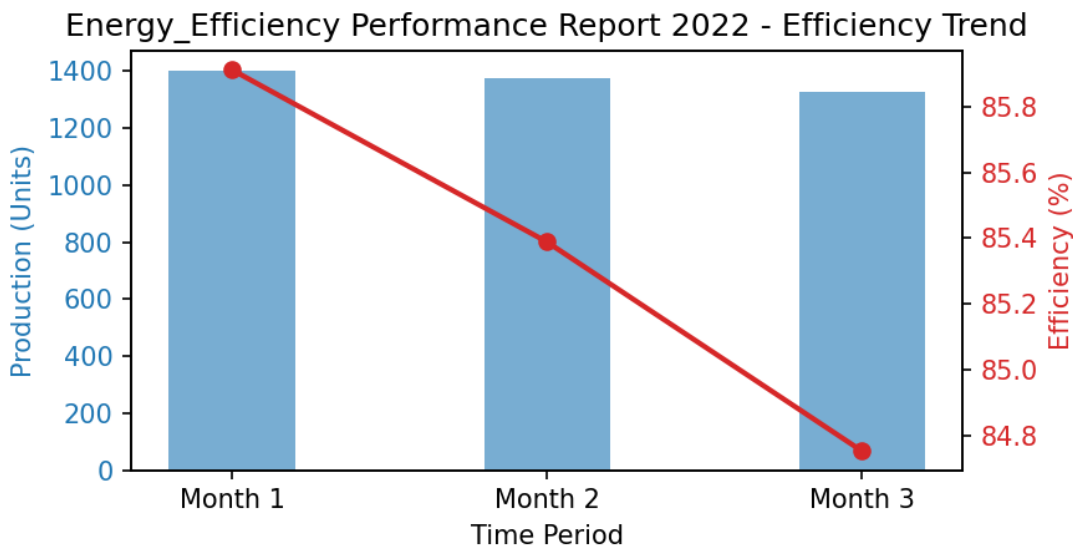
During the operational assessment of Energy_Efficiency Performance Report 2022, production efficiency experienced variance. The primary root cause for downtime in Q2 was unexpected hydraulic failure in Pump-04, whereas in Q4 the downtime was driven by supply chain shortages of raw titanium alloy.

2. Manufacturing Metrics & Output Table

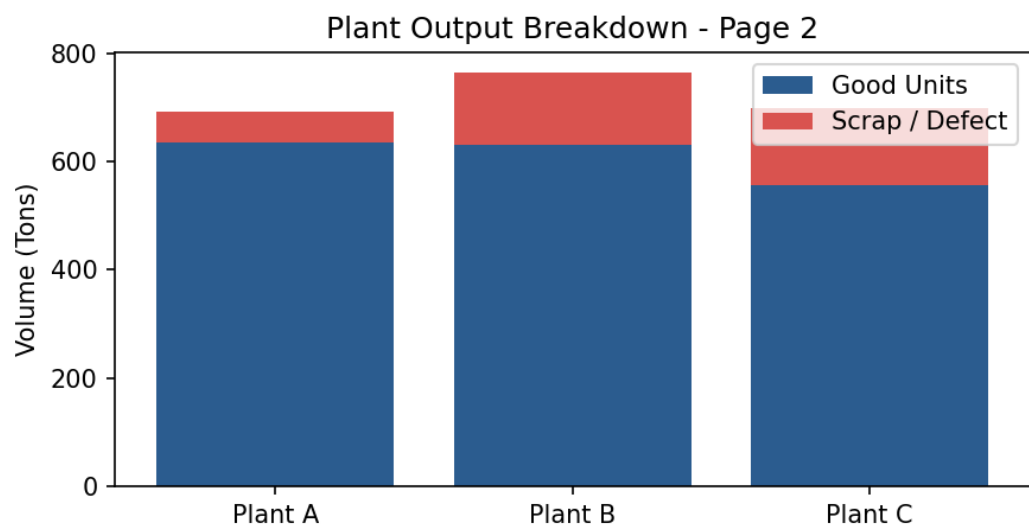
Metric Category	Quarter Baseline	Actual Output*	Efficiency Rate (%)
Line Alpha (Heavy)	1200 Units/Day	1178 Units/Day	86.1%
Line Beta (Light)	800 Units/Day	824 Units/Day	77.5%
Total Operational Output	2000 Units/Day	2002 Units/Day	81.8%

**Note: Actual output includes unadjusted scrap margin of 3.5% for Units/Day.*

3. Dual-Axis Production vs Efficiency Trend



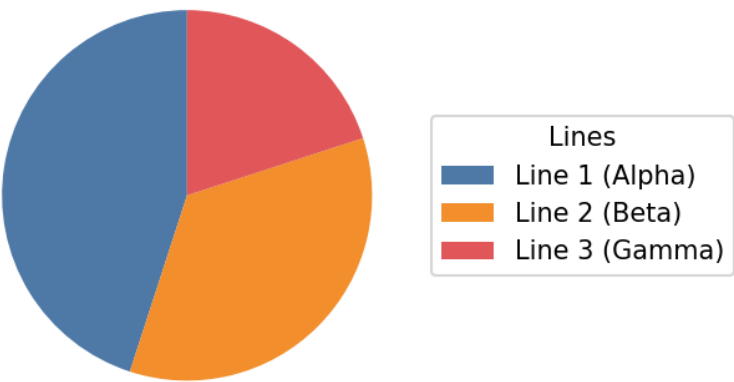
Page 2 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

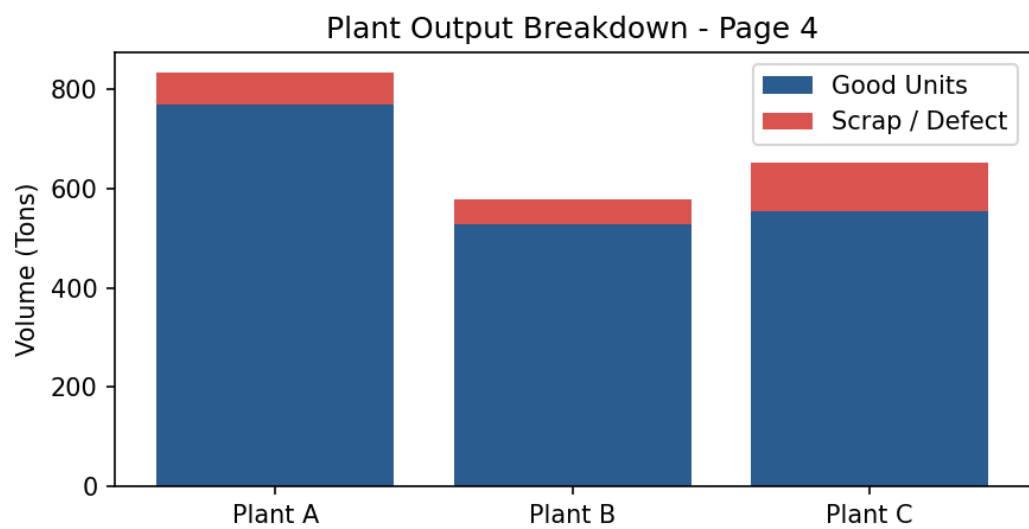
Page 3 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 3



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

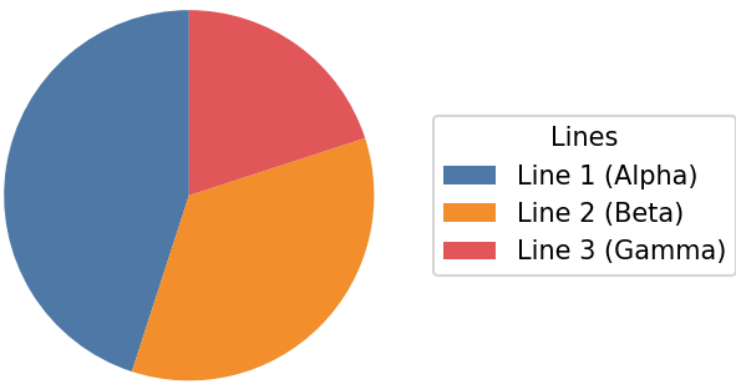
Page 4 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

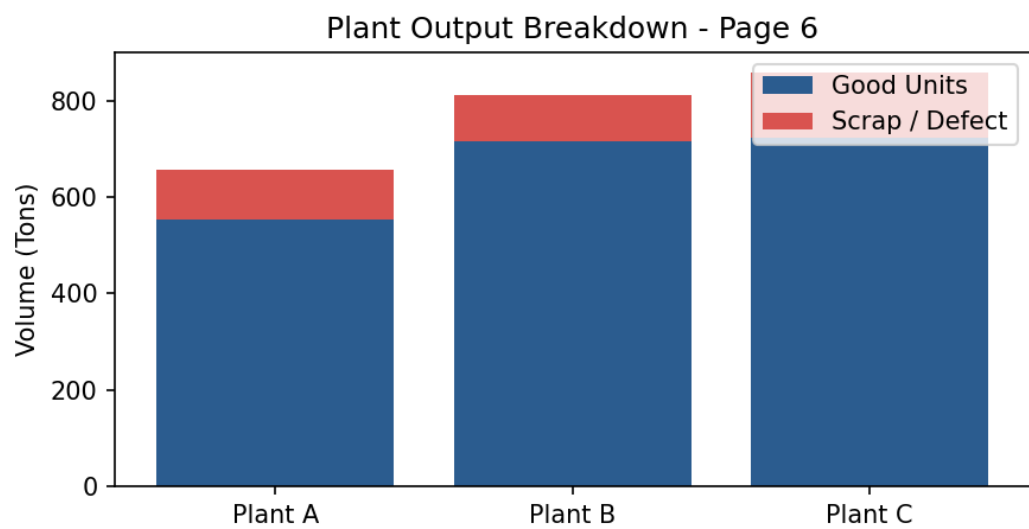
Page 5 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 5



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

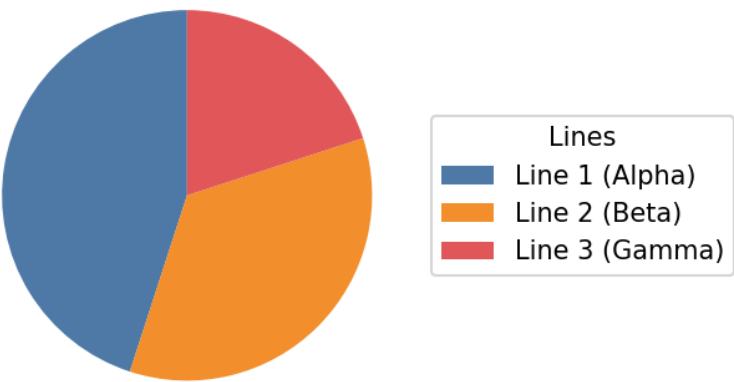
Page 6 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

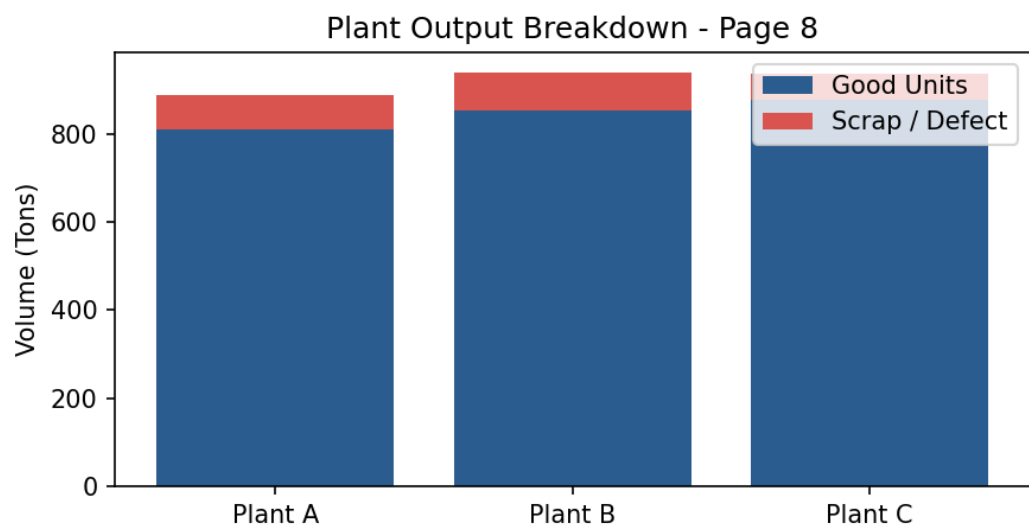
Page 7 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 7



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

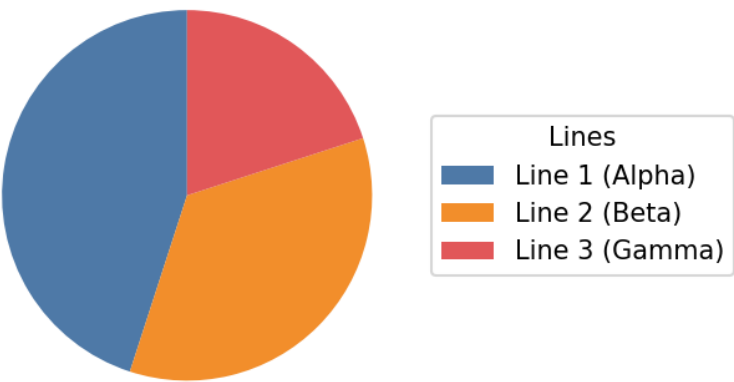
Page 8 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

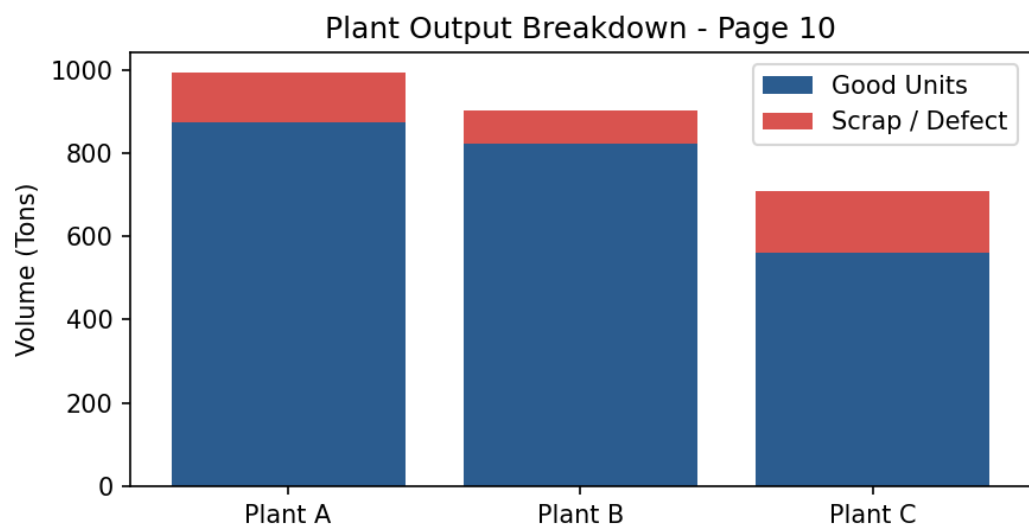
Page 9 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 9



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

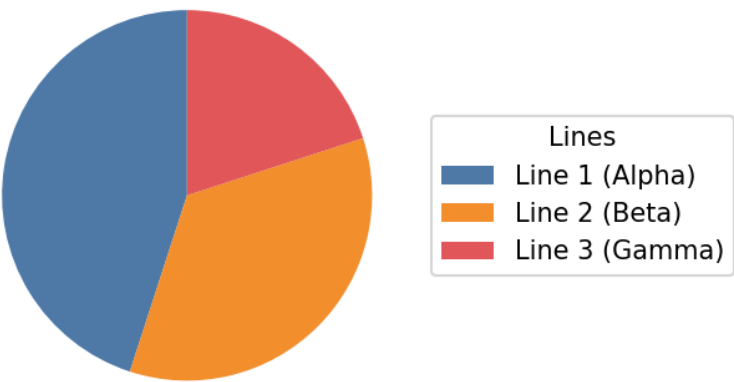
Page 10 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

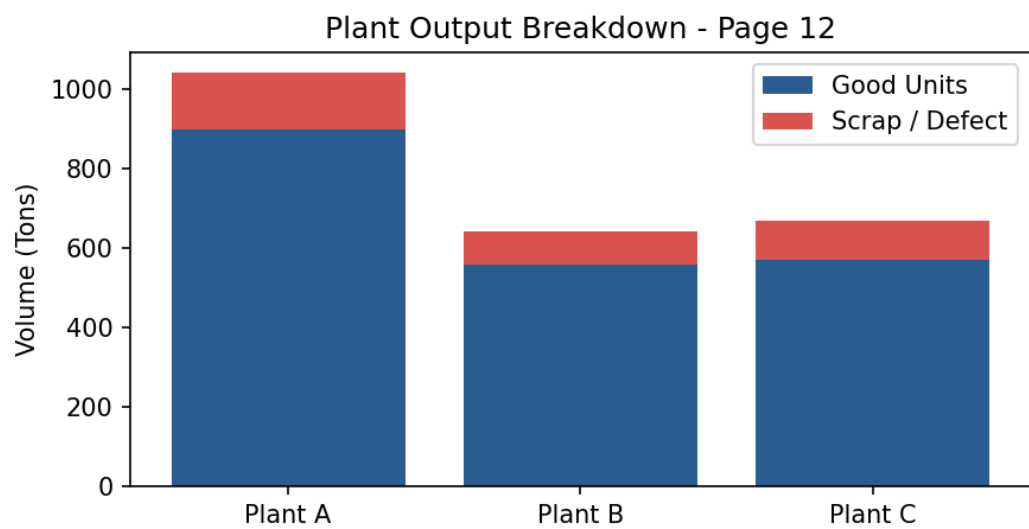
Page 11 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 11



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

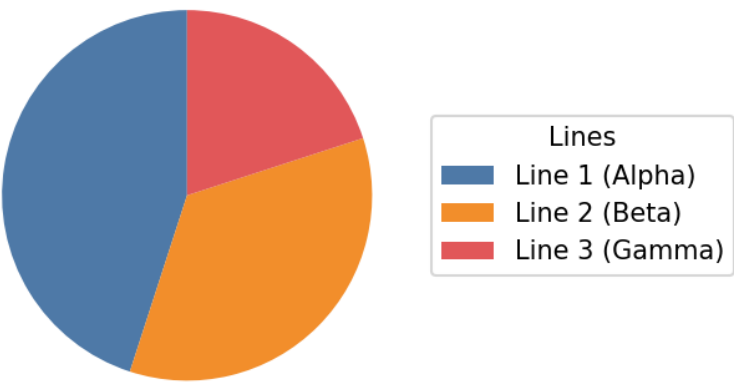
Page 12 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

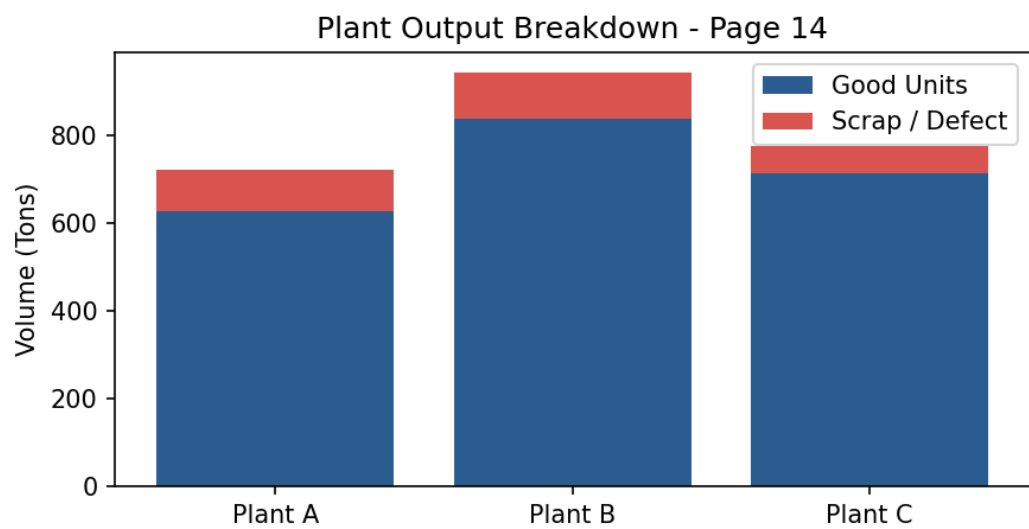
Page 13 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 13



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

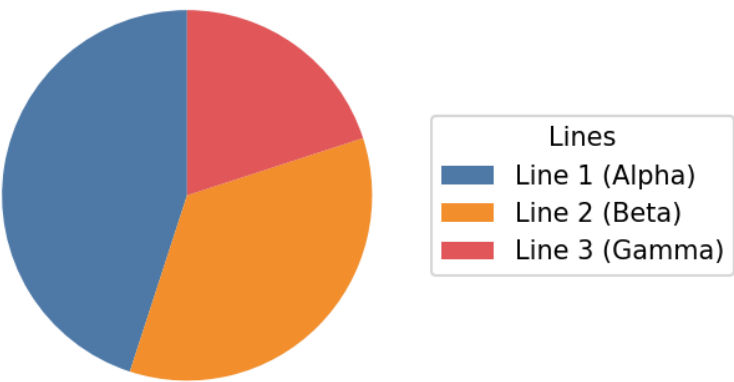
Page 14 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

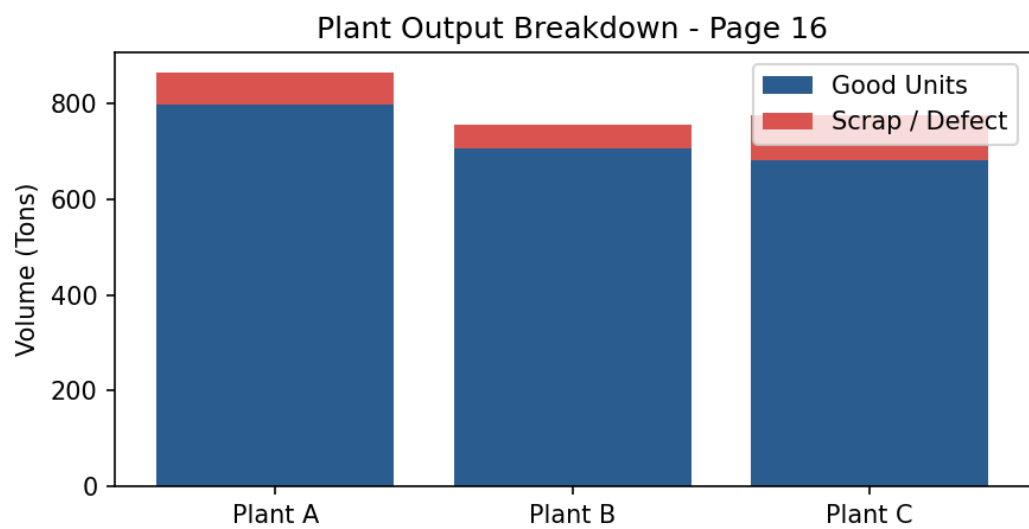
Page 15 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 15



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

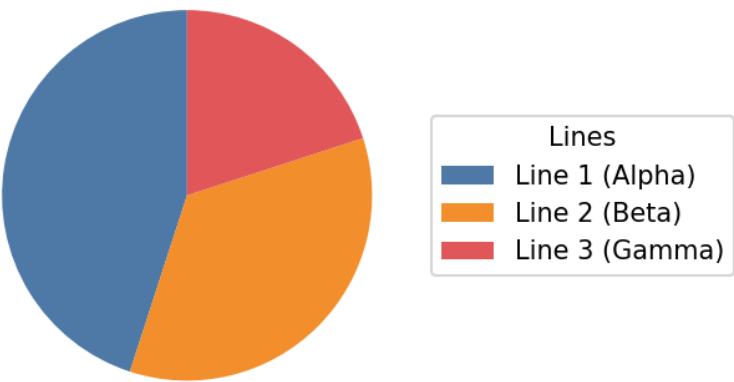
Page 16 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

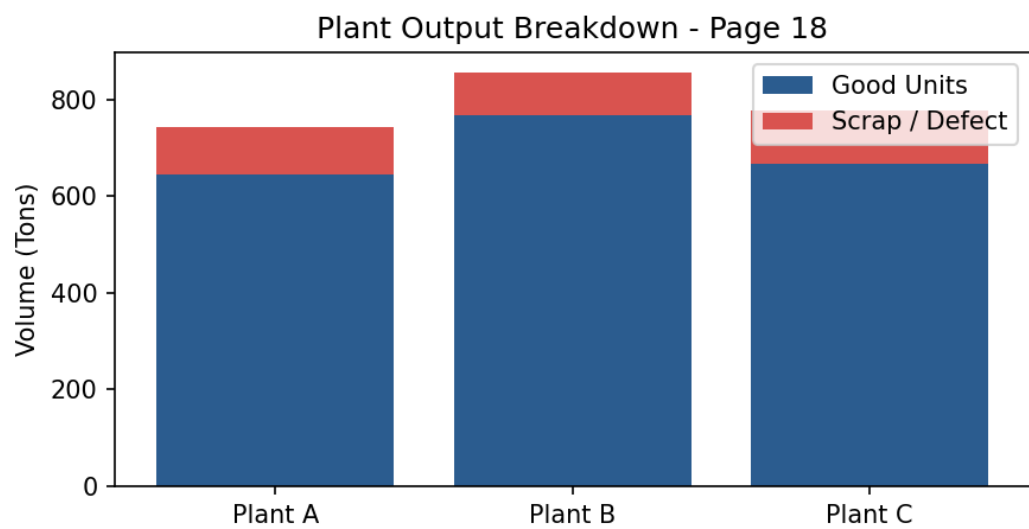
Page 17 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 17



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

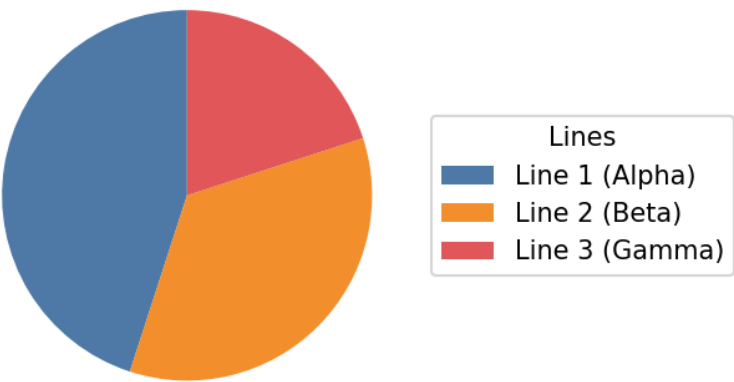
Page 18 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

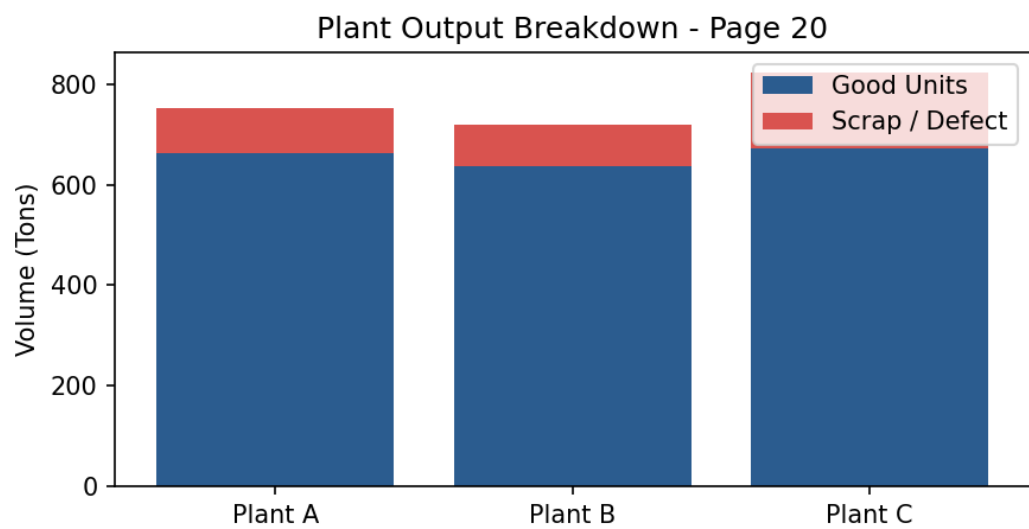
Page 19 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 19



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

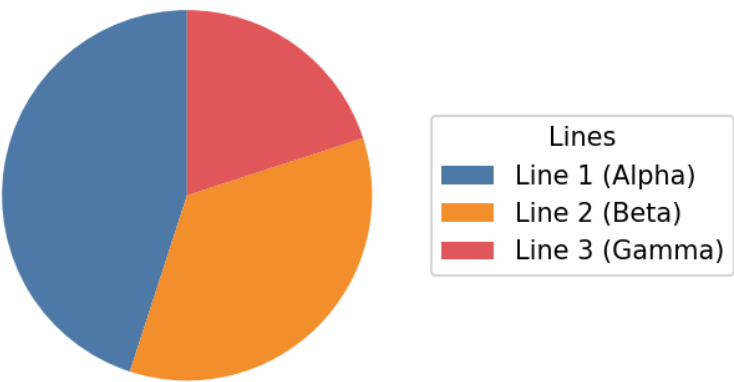
Page 20 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

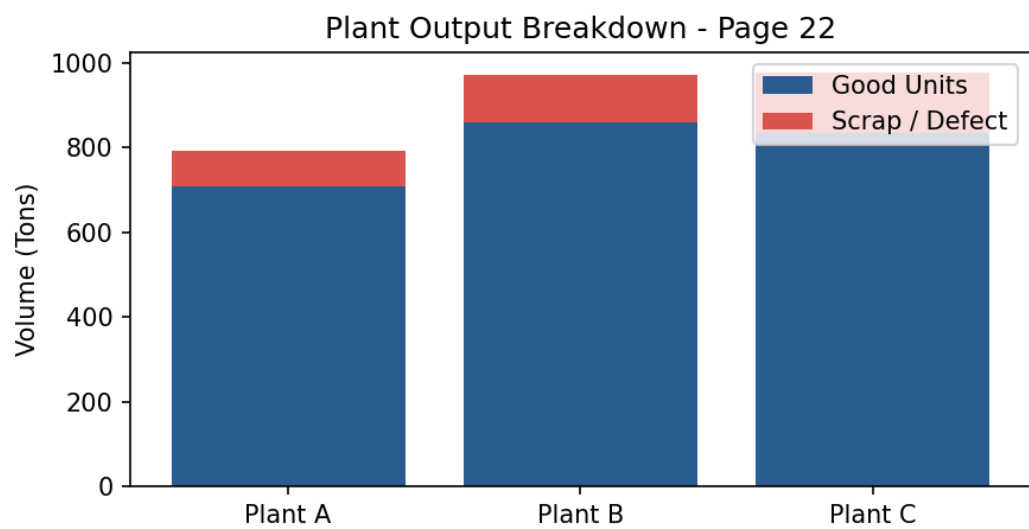
Page 21 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 21



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

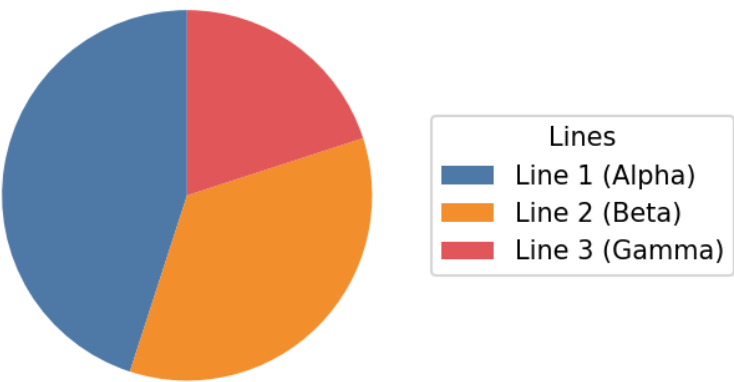
Page 22 - Extended Operations & Analysis



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS

Page 23 - Extended Operations & Analysis

Distribution Share (Legend Only) - Page 23



ID	Component	Spec Value	Compliance
C-001	Valve Subsystem 1	105.2 psi	PASS
C-002	Valve Subsystem 2	110.4 psi	PASS
C-003	Valve Subsystem 3	115.6 psi	PASS
C-004	Valve Subsystem 4	120.8 psi	PASS
C-005	Valve Subsystem 5	126.0 psi	PASS
C-006	Valve Subsystem 6	131.2 psi	PASS
C-007	Valve Subsystem 7	136.4 psi	PASS
C-008	Valve Subsystem 8	141.6 psi	PASS
C-009	Valve Subsystem 9	146.8 psi	PASS
C-010	Valve Subsystem 10	152.0 psi	PASS
C-011	Valve Subsystem 11	157.2 psi	PASS
C-012	Valve Subsystem 12	162.4 psi	PASS
C-013	Valve Subsystem 13	167.6 psi	PASS
C-014	Valve Subsystem 14	172.8 psi	PASS